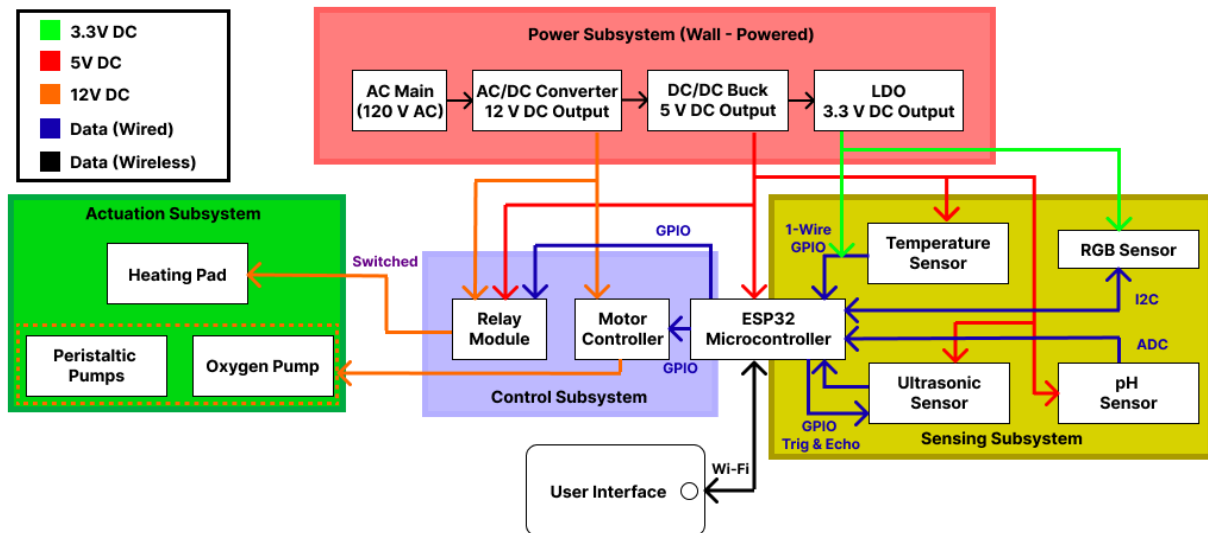


Kombucha Fermentation Control System

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Block Diagram:



Kombucha quality depends on precise control of temperature, acidity, and fermentation progression. These are parameters that traditional home brewing leaves to manual monitoring, making consistent results difficult and drift easy to miss. This system automates that process using an ESP32 that continuously reads temperature, pH, liquid level, and color state, driving a heating pad, dosing pumps, and oxygen pump to maintain user-defined targets, with all data streamed to a Wi-Fi dashboard for real-time visibility and timestamped logging.

High-Level Requirements:

- The system shall continuously monitor temperature, pH, color state (HSL), and liquid level, display these measurements on a Wi-Fi dashboard with timestamped logging, and provide user access to actuator status and control.
- For a demonstration volume of approximately 800 mL starting near room temperature, the system shall command heating and produce a measurable upward temperature trend toward a user-defined setpoint, demonstrated by at least $+1.0^{\circ}\text{C}$ change or entering the $\pm 1.0^{\circ}\text{C}$ band when starting within a few degrees of the setpoint
- The system shall actively modify liquid acidity using pump-driven dosing and shall produce a measurable decrease in pH of ≥ 1.5 change during demonstration testing.
- The system shall detect and report a measurable visual-state shift from an initially clear or lightly colored state to a visibly colored state using color-sensor-derived hue, saturation, and lightness (HSL) values during demonstration testing.

Requirements & Verification Table:

Subsystem	Requirement	Verification	Status
Power	5 V rail $\pm 5\%$	DMM measurement	Done
Power	3.3 V rail $\pm 5\%$	DMM measurement	Done
Power	12 V rail $\pm 10\%$ under full load (pump + heater active)	DMM under load	Done
Power	No reset during simultaneous pump + relay switching over 5 min	Observation/Serial log	Done
Sensing	Ultrasonic level ± 1 cm repeatability across all 3 jars	Serial log/dashboard	Done
Sensing	Temperature stable to $\pm 1^\circ\text{C}$	Serial log/dashboard	Done
Sensing	pH repeatability ± 0.3 over 5 min; detect ≥ 1.5 pH unit change	Serial log/dashboard	Done
Sensing	RGB/HSL color state updated; detects light-to-dark shift	Serial log/dashboard	Done
Sensing	All sensing channels valid simultaneously over 5 min	Serial log/dashboard	Done
Sensing	All ESP32 input signals remain within 0–3.6 V during operation	DMM measurement	Done
Control	Heater ON below temp range, OFF above; $\geq 0.15^\circ\text{C}/\text{min}$ for 10 min	Serial log/observe relay behavior	Done
Control	Liquid level maintained in the main jar via pump actuation	Observation	Done
Control	pH logged and available for review; no direct pH-triggered actuation required	Observation	Done
Control	Oxygen pump activated on schedule	Observation	Done
Control	All actuators default to OFF at boot/reset	Observation	Done
User	Displays real-time system measurements and actuator states	Observation	Done
User	Shows system alerts, errors, and operator actions	Observation	Done